Package 'DegradSim'

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Title Simulation of Lévy and Non-Homogeneous Lévy Degradation Processes

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Description Exact simulation, at given inspection times, of one or several degradation trajectories of Lévy processes (gamma, Wiener, inverse Gaussian, variance-gamma and Meixner processes) and of non-homogeneous Lévy processes obtained by a time change X(t) = Y(A(t)) of a gamma, Wiener or inverse Gaussian process (Kahle, Mercier and Paroissin, 2016, ISBN:978-1-84821-888-8; Cha and Mercier, 2019, <doi:10.1017/apr.2019.21>); inspection times with a chosen pattern, plotting of the trajectories and extraction of the simulated data.

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URL https://github.com/rytracetito/levy_packages/tree/main/DegradSim, https://github.com/rytracetito/levy_packages

BugReports https://github.com/rytracetito/levy_packages/issues

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Contents

DegradSim-package 2
inspection_times 6
plot_plan 8
plot_traj 9
sim_gamma 10
sim_ig 13
sim_meixner 16

sim_vg 19
sim_wiener 21
traj_df 24

Index **26**

`DegradSim-package` *DegradSim: simulation of Lévy and non-homogeneous Lévy degradation processes*

Description

`DegradSim` simulates exactly, at given inspection times, one or several degradation trajectories of two families of processes:

1. **Lévy processes** (homogeneous case): gamma, Wiener, inverse Gaussian, variance-gamma and Meixner processes;
2. **non-homogeneous Lévy processes** obtained by a time change $X(t) = Y(A(t))$ of a Lévy process Y: gamma, Wiener and inverse Gaussian processes.

Details

The inspection times can be irregular and differ from one trajectory to another; `inspection_times()` builds them with a chosen pattern. The simulated trajectories can be printed, plotted and extracted as a data frame.

Lévy processes

Definition (Sato, 2001, p. 3). A real stochastic process $(X_t)_{t\geq 0}$ is a Lévy process if

1. it has independent increments: for all $n \geq 1$ and $0 \leq t_0 < t_1 < \cdots < t_n$, the random variables $X_{t_0}, X_{t_1} - X_{t_0}, \ldots, X_{t_n} - X_{t_{n-1}}$ are independent;
2. it starts at the origin: $X_0 = 0$ almost surely;
3. it is time homogeneous: the distribution of $(X_{s+t} - X_s)_{t\geq 0}$ does not depend on s;
4. it is stochastically continuous: for all $\varepsilon > 0$, $P(|X_{s+t} - X_s| > \varepsilon) \to 0$ as $t \to 0$;
5. its trajectories $t \mapsto X_t$ are right-continuous with left limits (càdlàg), almost surely.

Infinitely divisible distribution (Sato, 2001, p. 4). A distribution μ on $\mathbb{R}$ is infinitely divisible if, for every integer $n \geq 1$, there is a distribution μ_n such that $\mu = \mu_n^{n*}$, the n-fold convolution of μ_n: a random variable with distribution μ has the distribution of the sum of n independent random variables with distribution μ_n. For a Lévy process, the distribution of X_t is infinitely divisible for every t; conversely, every infinitely divisible distribution is the distribution of X_1 for a Lévy process, unique in law (Sato, 2001, Theorems 1.1 and 1.2). The gamma, normal, inverse Gaussian, variance-gamma and Meixner distributions are infinitely divisible.

The simulation functions, with the distribution of an increment over a time $\Delta t = t - s$:

Function	Process	Parameters	$X(t) - X(s)$
`sim_gamma()`	gamma	`a`, `b`	$\Gamma(a\Delta t, b)$, rate b
`sim_wiener()`	Wiener with drift	`mu`, `sigma`	$N(\mu\Delta t, \sigma^2\Delta t)$
`sim_ig()`	inverse Gaussian	`mu`, `eta`	$IG(\mu\Delta t, \eta\Delta t^2)$
`sim_vg()`	variance-gamma	`theta`, `sigma`, `nu`	$VG(\theta, \sigma, \nu)$ over Δt
`sim_meixner()`	Meixner	`a`, `b`, `d`, `m`	$Meixner(a, b, d\Delta t, m\Delta t)$

Non-homogeneous Lévy processes

Time change. Let $A : [0, +\infty[\to [0, +\infty[$ be a non-decreasing function, right-continuous with left limits, with $A(0) = 0$ and $A(t) \to +\infty$ as $t \to +\infty$, and let Y be a homogeneous Lévy process. The process

$$X(t) = Y(A(t)), \qquad t \geq 0,$$

is a non-homogeneous Lévy process (Kahle et al., 2016, p. xix-xx; Cha and Mercier, 2019, section 2): $X(0) = 0$, its increments are independent and $X(t) - X(s)$ has the distribution of $Y(A(t) - A(s))$. Its increments are not stationary when A is not linear. When A is continuous, X is also stochastically continuous: it is an additive process in the sense of Sato (2001, p. 3), that is a process satisfying the conditions 1, 2, 4 and 5 above.

Identifiability. The time change A and one parameter of Y cannot be identified separately: replacing A by cA can be compensated by a change of this parameter. This parameter is therefore set to 1 and contained in A. With $\Delta A = A(t) - A(s)$:

Function	Parameters	$X(t) - X(s)$	Fixed	Homogeneous case
`sim_gamma()`	`b`, `A`	$\Gamma(\Delta A, b)$	$a = 1$	$A(t) = at$
`sim_wiener()`	`sigma`, `A`	$N(\Delta A, \sigma^2 \Delta A)$	$\mu = 1$	$A(t) = \mu t$, $\sigma/\sqrt{\mu}$
`sim_ig()`	`eta`, `A`	$IG(\Delta A, \eta \Delta A^2)$	$\mu = 1$	$A(t) = \mu t$, η/μ^2

The time-scaled Wiener process is $X(t) = A(t) + \sigma B(A(t))$, with B a standard Brownian motion. The last column gives the time change (and the parameter) that give back the homogeneous process of the first table. In each function, the argument `A` is `NULL` by default (homogeneous process); a function `A` gives the non-homogeneous process, and the fixed parameter must then not be given. The variance-gamma and Meixner processes are simulated in the homogeneous case only.

Getting started

A simulation always follows the same three steps.

1. **Choose the inspection times** `t`. Three forms are accepted by every `sim_` function:
   - a numeric vector, for instance `seq(0, 10, by = 0.5)`: all the trajectories share these times, and their number is given by `n_traj`;
   - a list of numeric vectors, one per trajectory, for instance `list(c(0, 1, 3), c(0, 2, 4, 6))`;
   - the result of `inspection_times()`, for instance with the pattern `"increasing"` for trajectories inspected more and more often.

   The times of each trajectory must be finite, start at 0 and be strictly increasing, with at least two times (0 and one inspection). These conditions are checked by every `sim_` function, also for the result of `inspection_times()`, and an error explains which condition fails.
2. **Simulate** with the function of the chosen process and its parameters, for instance `sim_gamma(t, a = 2, b = 3)` for a homogeneous gamma process. For a non-homogeneous process, give the time change instead of the fixed parameter, for instance `sim_gamma(t, b = 3, A = function(t) 2 * t^1.5)`. The function `A` must be vectorized, with $A(0) = 0$, finite and non-decreasing at the inspection times: this is checked too. The conditions "right-continuous with left limits" and $A(t) \to +\infty$ cannot be checked on a finite set of times and are left to the user.
3. **Store and use the result** `x`, an object of class `DegradSim`. Its form depends on the times:
   - same times for all the trajectories: a matrix with the column `t` and one column per trajectory, `X1`, ..., `Xn`; the levels of the k-th trajectory are `x[, k + 1]` or `x[, "Xk"]`;

- different times: a list with one data frame (`t`, `X`) per trajectory; the k-th trajectory is `x[[k]]`, with times `x[[k]]$t` (or `x[[k]][, 1]`) and levels `x[[k]]$X` (or `x[[k]][, 2]`).

In both cases, `x` prints the model, the parameters, the time change if any and the first simulated values; `traj_df(x)` gives all the data in one data frame (`id`, `t`, `X`, and `dt`, `dX` with `increments = TRUE`) and `plot_traj(x)` plots the trajectories.

Use `set.seed()` before a simulation to make it reproducible.

Other functions:

- `inspection_times()`: inspection times following a chosen pattern (regular, increasing, decreasing, or alternating between frequent and rare inspections);
- `plot_plan()`: plot of the inspection times, one row per trajectory (`ggplot2`);
- `plot_traj()`: plot of the simulated trajectories (`ggplot2`);
- `traj_df()`: simulated data as a data frame.

Exact simulation at the inspection times (random walk approximation)

All the simulated processes have independent increments with known distributions. The values at the inspection times $0 = t_0 < t_1 < \cdots < t_n$ are therefore simulated by the random walk approximation (RWA) of Kahle et al. (2016, algorithms 1.7 and 2.32): independent increments ΔX_i with the distribution of $X(t_i) - X(t_{i-1})$ are drawn, then $X(t_0) = 0$ and $X(t_i) = \Delta X_1 + \cdots + \Delta X_i$. The vector $(X(t_1), \ldots, X(t_n))$ has exactly the distribution of the process at the inspection times; the name "approximation" refers only to the path between two inspection times, which is not simulated. The increments are drawn with:

- gamma: `stats::rgamma()`; Wiener: `stats::rnorm()`;
- inverse Gaussian: method of Michael, Schucany and Haas (1976);
- variance-gamma: subordination of a Brownian motion by a gamma process, or difference of two gamma processes (argument `method`);
- Meixner: numerical inversion of the distribution function (method PINV of `Runuran`, error of order 10^{-10}).

The details are given in the help page of each function.

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See Also

Useful links:

- https://github.com/rytracetito/levy_packages/tree/main/DegradSim
- https://github.com/rytracetito/levy_packages
- Report bugs at https://github.com/rytracetito/levy_packages/issues

Examples

```
# 1. Inspection times: 10 trajectories, inspected more and more often
plan <- inspection_times("increasing", n = 20, n_traj = 10, tmax = 100)
plot_plan(plan)

# 2. Simulation of a homogeneous gamma process (a Lévy process)
set.seed(1)
x <- sim_gamma(plan, a = 0.2, b = 1)

# 3. Use of the result: the same times for all the trajectories give a
#    matrix (column t, then X1, ..., X10)
x
x[, "X1"]
head(traj_df(x, increments = TRUE))
plot_traj(x)

# 4. Non-homogeneous gamma process with the time change A(t) = 0.02 t^1.5
y <- sim_gamma(plan, b = 1, A = function(t) 0.02 * t^1.5)
y
plot_traj(y, "Non-homogeneous gamma process")

# 5. Different inspection times: a list of data frames (t, X)
z <- sim_wiener(list(c(0, 1, 3, 6), c(0, 2, 4, 6, 8, 10)), mu = 0.5, sigma = 1.5)
z[[2]]
z[[2]]$X
```

`inspection_times` *Inspection times following a chosen pattern*

Description

Generates inspection times on $[0, t_{max}]$ for one or several trajectories, with gaps between two successive inspections that are constant, increasing, decreasing, or that vary between the beginning, the middle and the end of the period. The result can be given directly to the simulation functions, such as `sim_gamma()`, and plotted with `plot_plan()`.

Usage

```
inspection_times(
  pattern = "regular",
  n = 10,
  n_traj = 1,
  tmax = 100,
  digits = NULL,
  a = 0.5
)
```

Arguments

`pattern`	Name of the pattern, `"regular"` (default), `"increasing"`, `"decreasing"`, `"inc-dec"` or `"dec-inc"` (see Details), or a strictly increasing R function on $[0, 10]$.
`n`	Number of inspections after time 0, a positive integer. A single value gives the same number of inspections to all the trajectories; a vector of length `n_traj` gives one number per trajectory.
`n_traj`	Number of trajectories, a positive integer.
`tmax`	Time of the last inspection, $t_{max} > 0$. A single value gives the same last time to all the trajectories; a vector of length `n_traj` gives one value per trajectory.
`digits`	Number of decimal places of the times, a non-negative integer, or `NULL` (default) for no rounding. Times made equal by the rounding are removed, with a warning (see Details).
`a`	Amplitude of the patterns `"inc-dec"` and `"dec-inc"`, in $[0, 1[$: it sets the contrast between frequent and rare inspections, from 0 (regular inspections) to 1 excluded; default 0.5. Not used by the other patterns.

Details

Construction. The times are the values of a strictly increasing function f on the regular grid $x_i = i\, x_{max}/n$, $i = 0, \ldots, n$, of $[0, x_{max}]$ with $x_{max} = 10$, rescaled to $[0, t_{max}]$:

$$t_i = t_{max} \frac{f(x_i) - f(0)}{f(x_{max}) - f(0)}, \qquad i = 0, \ldots, n.$$

So $t_0 = 0$, $t_n = t_{max}$, and the gap between two successive inspections is about proportional to the slope $f'(x_i)$: the inspections are frequent where f is flat and rare where f is steep.

Why the interval $[0, 10]$. The rescaling removes the scale of the values of f, but not the scale of x: for a non-linear f, the length of the interval sets the contrast between the smallest and the largest

gap. For $f(x) = \ln(x+1)$, the ratio of the first gap to the last one is about $f'(0)/f'(x_{max}) = x_{max} + 1$: 2 on $[0, 1]$ (almost regular inspections), 11 on $[0, 10]$ and 101 on $[0, 100]$ (almost all the inspections at one end). The value $x_{max} = 10$ gives a clear but moderate contrast. It is fixed, and a function given as `pattern` is also evaluated on $[0, 10]$.

Patterns.

- `"regular"`: $f(x) = x$, constant gaps t_{max}/n;
- `"increasing"`: $f(x) = \ln(x+1)$, decreasing gaps: inspections rare at the beginning and more and more frequent;
- `"decreasing"`: $f(x) = -\ln(x_{max} + 1 - x)$, increasing gaps: inspections frequent at the beginning and rarer and rarer, mirror image of `"increasing"`;
- `"inc-dec"`: $f(x) = x + a\,\frac{x_{max}}{2\pi}\sin(2\pi x/x_{max})$, gaps that decrease then increase: inspections rare at the beginning, frequent in the middle, rare at the end;
- `"dec-inc"`: $f(x) = x - a\,\frac{x_{max}}{2\pi}\sin(2\pi x/x_{max})$, gaps that increase then decrease: inspections frequent at the beginning, rare in the middle, frequent at the end.

The names describe the number of inspections per unit of time: `"increasing"` means more and more inspections, `"inc-dec"` more, then fewer inspections.

Amplitude `a` of `"inc-dec"` and `"dec-inc"`. Here $f'(x) = 1 \pm a\cos(2\pi x/x_{max})$: the gaps vary between $1-a$ and $1+a$ times the mean gap t_{max}/n, so that the largest gap is $(1+a)/(1-a)$ times the smallest one. With the default $a = 0.5$, this ratio is 3 and the gaps change progressively; with $a = 0.8$, it is 9 and the frequent inspections gather in a short time; $a = 0$ gives regular inspections.

Rounding. With `digits`, the times are rounded to `digits` decimal places. If the rounding makes two successive times equal, the repeated times are removed, so that the trajectory has fewer than `n` inspections after time 0, and a warning gives the number of inspections that are kept.

Other patterns. Any strictly increasing R function on $[0, 10]$ can be given as `pattern`, for instance `sqrt`; an error is raised if it is not strictly increasing on the grid.

Value

A data frame with one row per inspection and two columns: `id` (number of the trajectory, 1 to `n_traj`) and `time` (inspection time). Each trajectory starts at time 0 and ends at its `tmax`, with strictly increasing times. The times of the trajectory k are `plan$time[plan$id == k]`.

See Also

`plot_plan()` to plot the inspection times, `sim_gamma()` and the other simulation functions, which accept the result as inspection times.

Examples

```
# 1. Regular inspections: 10 inspections after time 0, up to 100
inspection_times("regular", n = 10, tmax = 100)

# 2. More and more frequent inspections, times rounded to integers
inspection_times("increasing", n = 10, digits = 0)

# 3. Fewer and fewer inspections
inspection_times("decreasing", n = 10, digits = 0)

# 4. Rare at the beginning, frequent in the middle, rare at the end
inspection_times("inc-dec", n = 20, digits = 1)
```

```
# 5. Frequent at the beginning and at the end, rare in the middle
inspection_times("dec-inc", n = 20, digits = 1)

# 6. Stronger contrast between frequent and rare inspections
inspection_times("dec-inc", n = 20, digits = 1, a = 0.8)

# Gaps between inspections: from about 2.5 to 7.5 with the default amplitude
range(diff(inspection_times("dec-inc", n = 20)$time))

# 7. Three trajectories with 10, 15 and 20 inspections, up to 50, 100 and 150
plan <- inspection_times("increasing", n = c(10, 15, 20), n_traj = 3,
                         tmax = c(50, 100, 150), digits = 1)
plan

# Times of the second trajectory only
plan$time[plan$id == 2]

# 8. Same number of inspections and same tmax: the same times for the
#    three trajectories
inspection_times("increasing", n = 5, n_traj = 3, tmax = 10)

# 9. Rounding too strong: the repeated times are removed with a warning
inspection_times("decreasing", n = 30, tmax = 20, digits = 0)

# 10. Any strictly increasing function on [0, 10]: with sqrt, the
#     inspections are rare at the beginning and more and more frequent
inspection_times(sqrt, n = 10, digits = 0)

# 11. Mixture of the increasing and decreasing patterns: frequent
#     inspections around a point that moves to the right when w increases
#     (0 < w < 1)
w <- 0.8
inspection_times(function(x) w * log(x + 1) - (1 - w) * log(11 - x), n = 20, digits = 1)

# 12. Plot the inspection times, one row per trajectory, then simulate
plan <- inspection_times("dec-inc", n = c(10, 20, 30), n_traj = 3, tmax = 100)
plot_plan(plan)
x <- sim_gamma(plan, a = 0.2, b = 1)
plot_traj(x)
```

`plot_plan` *Plot inspection times*

Description

Draws the inspection times of one or several trajectories with `ggplot2`: one point per inspection, and one row per trajectory when there are several. It shows at a glance where the inspections are frequent and where they are rare.

Usage

```
plot_plan(t, title = "Inspection times")
```

Arguments

t	Inspection times, in one of the three forms accepted by the sim_ functions: • a numeric vector, for one trajectory, for instance seq(0, 10, by = 0.5); • a list of numeric vectors, one per trajectory, for instance list(c(0, 1, 3), c(0, 2, 4, 6)); • the result of inspection_times(), one plan per trajectory. The times of each trajectory start at 0 and are strictly increasing.
title	Title of the plot, a character string; by default "Inspection times".

Value

A ggplot object. It is displayed when printed, and it can be modified with the usual ggplot2 functions, added with +: for instance + ggplot2::labs(x = "Time (hours)") to change the axis label.

See Also

inspection_times(), plot_traj()

Examples

```
# One trajectory inspected every 0.5 time unit
plot_plan(seq(0, 10, by = 0.5))

# Inspections more and more frequent over time, with a custom title
t <- inspection_times("increasing", n = 30, tmax = 100)
plot_plan(t, "Increasing number of inspections")

# Five trajectories, each with its own times: one row per trajectory
plan <- inspection_times("dec-inc", n = c(10, 15, 20, 25, 30), n_traj = 5, tmax = 100)
plot_plan(plan)

# Different inspection times given as a list
plot_plan(list(c(0, 1, 3, 6), c(0, 2, 4, 6, 8, 10)))

# Change the axis label
plot_plan(t) + ggplot2::labs(x = "Time (hours)")
```

---

plot_traj *Plot simulated trajectories*

Description

Draws the trajectories simulated by a sim_ function with ggplot2: one grey line per trajectory, with a point at each inspection.

Usage

```
plot_traj(x, title = NULL)
```

Arguments

x	Result of a simulation function of the package, for instance sim_gamma().
title	Title of the plot, a character string; by default the name of the process, for instance "gamma process".

Value

A ggplot object. It is displayed when printed, and it can be modified with the usual ggplot2 functions, added with +: for instance + ggplot2::geom_hline(yintercept = 10) to draw a failure threshold, or + ggplot2::labs(x = "Time (hours)") to change an axis label.

See Also

traj_df()

Examples

```
set.seed(1)
x <- sim_wiener(seq(0, 10, by = 0.5), mu = 0.5, sigma = 1.5, n_traj = 5)
plot_traj(x)

# Custom title
plot_traj(x, "Five trajectories of a Wiener process")

# Add a failure threshold and change the axis labels
plot_traj(x) +
  ggplot2::geom_hline(yintercept = 5, linetype = "dashed", colour = "red") +
  ggplot2::labs(x = "Time (hours)", y = "Degradation (%)")
```

---

sim_gamma *Simulation of the gamma process*

Description

Simulates exactly, at given inspection times, one or several trajectories of a homogeneous gamma process (a Lévy process) or of a non-homogeneous gamma process, obtained by a time change A.

Usage

```
sim_gamma(t, a, b, A = NULL, n_traj = 1)
```

Arguments

t	Inspection times. The times of each trajectory must start at 0 and be strictly increasing. Three forms are accepted: • a numeric vector, for instance seq(0, 10, by = 0.5): all the trajectories share these times, and their number is n_traj; • a list of vectors, one per trajectory, for instance list(c(0, 1, 3), c(0, 2, 4, 6)): each trajectory has its own times (n_traj is then ignored); • the data frame returned by inspection_times(), which gives one plan per trajectory (n_traj is then ignored).

a	Shape parameter of the homogeneous process, $a > 0$: the increment over a time Δt has shape $a\Delta t$. Not given when A is given (the shape is then $A(t) - A(s)$).
b	Rate parameter, $b > 0$ (the scale is $1/b$).
A	NULL (default) for the homogeneous process, or the time change for the non-homogeneous process: a vectorized R function of t, non-decreasing with $A(0) = 0$, for instance function(t) 2 * t^1.5.
n_traj	Number of trajectories to simulate when t is a vector (default 1), a positive integer.

Value

An object of class DegradSim, whose form depends on the inspection times:

- **same times for all the trajectories** (t a vector, or a plan of inspection_times() with the same times for all the trajectories): a matrix with columns t (inspection times), X1, ..., Xn (simulated degradation levels of the trajectories 1 to n, equal to 0 at t = 0). The levels of the k-th trajectory are x[, k + 1] or x[, "Xk"];
- **different times** (t a list, or a plan of inspection_times() with different times): a list with one data frame per trajectory, with columns t and X. The k-th trajectory is the data frame x[[k]], its times are x[[k]]$t (or x[[k]][, 1]) and its levels x[[k]]$X (or x[[k]][, 2]).

The model, the parameters and, for a non-homogeneous process, the time change are stored as attributes. Printing the object shows them with the first simulated values. Use traj_df() to gather the data in a single data frame (with the increments if needed) and plot_traj() to plot the trajectories.

Model

$\Gamma(k, b)$ denotes the gamma distribution with shape $k > 0$ and **rate** $b > 0$ (scale $1/b$), of density $b^k x^{k-1} e^{-bx}/\Gamma(k)$, $x > 0$.

Homogeneous gamma process (A = NULL, default). Let $a > 0$ and $b > 0$. The process satisfies $X(0) = 0$, has independent increments, and for $0 \leq s < t$,

$$X(t) - X(s) \sim \Gamma\big(a(t-s),\ b\big).$$

Then $E[X(t)] = at/b$ and $Var(X(t)) = at/b^2$. It is a Lévy process, increasing and purely made of jumps.

Non-homogeneous gamma process (A given). Let A be a time change: a non-decreasing function, right-continuous with left limits, with $A(0) = 0$ and $A(t) \to +\infty$ as $t \to +\infty$. The process is $X(t) = Y(A(t))$, where Y is the homogeneous gamma process with $a = 1$ and rate b, so that

$$X(t) - X(s) \sim \Gamma\big(A(t) - A(s),\ b\big),$$

$E[X(t)] = A(t)/b$ and $Var(X(t)) = A(t)/b^2$. This is the gamma process with shape function A and rate b of Kahle et al. (2016, chapter 2) and of Cha and Mercier (2019, Example 2). A classical choice is $A(t) = c\,t^\beta$.

Why $a = 1$ in the non-homogeneous case. With a shape $a\,(A(t) - A(s))$, the functions A and cA with the shapes a and a/c would give the same process: a and A could not be identified separately. The shape parameter is therefore contained in A, and the homogeneous process is the special case $A(t) = at$, which gives $\Gamma(a(t-s), b)$.

Exact simulation at the inspection times (random walk approximation)

The increments over disjoint intervals are independent with known distributions, so the values at the inspection times $0 = t_0 < t_1 < \cdots < t_n$ are simulated exactly by the random walk approximation (RWA) of Kahle et al. (2016, section 2.5.1, algorithm 2.32):

1. $\Delta_i = a(t_i - t_{i-1})$ (homogeneous case) or $\Delta_i = A(t_i) - A(t_{i-1})$ (non-homogeneous case), $i = 1, \ldots, n$;
2. draw independently $\Delta X_i \sim \Gamma(\Delta_i, b)$ with `stats::rgamma()`;
3. $X(t_0) = 0$ and $X(t_i) = \sum_{j \le i} \Delta X_j$.

The values at the inspection times have exactly the distribution of the process; only the path between two inspection times is not simulated. If A is constant between two inspection times, the increment is 0.

Conditions on the inspection times and on A

The times of each trajectory must be finite, start at 0 and be strictly increasing, with at least two times (0 and one inspection). The function `A` must be vectorized (`A(t)` returns one value per time), with $A(0) = 0$, finite and non-decreasing at the inspection times: these conditions are checked and an error is raised otherwise. The conditions "right-continuous with left limits" and $A(t) \to +\infty$ cannot be checked on a finite set of times: they are left to the user.

References

Kahle, W., Mercier, S. and Paroissin, C. (2016). *Degradation Processes in Reliability*. ISTE and Wiley, London. ISBN 978-1-84821-888-8.

Cha, J. H. and Mercier, S. (2019). Transformed Lévy processes as state-dependent wear models. *Advances in Applied Probability*, 51(2), 468-486. doi:10.1017/apr.2019.21

van Noortwijk, J. M. (2009). A survey of the application of gamma processes in maintenance. *Reliability Engineering & System Safety*, 94(1), 2-21. doi:10.1016/j.ress.2007.03.019

See Also

`sim_ig()` and `sim_wiener()` for the other processes with a time change, `plot_traj()`, `traj_df()`, `inspection_times()`, `plot_plan()`

Examples

```
# 1. Homogeneous process: one trajectory inspected every 0.5 time unit
t <- seq(0, 10, by = 0.5)
set.seed(1)
x <- sim_gamma(t, a = 2, b = 3)
x
plot_traj(x)

# 2. Ten trajectories with the same inspection times
x <- sim_gamma(t, a = 2, b = 3, n_traj = 10)
plot_traj(x)

# Same times for all the trajectories: a matrix with the column t,
# then one column per trajectory (X1, X2, ...)
head(x[, 1:4])
x[, "X2"]
x[, 3]
```

```
# 3. Irregular inspections, more and more frequent over time
plan <- inspection_times("increasing", n = 20, n_traj = 5, tmax = 10)
x <- sim_gamma(plan, a = 2, b = 3)
plot_traj(x)

# 4. Different inspection times for each trajectory: a list of data frames
x <- sim_gamma(list(c(0, 1, 3, 6), c(0, 2, 4, 6, 8, 10)), a = 2, b = 3)
x[[2]]
x[[2]]$X
x[[2]][, 2]

# 5. Non-homogeneous process with the time change A(t) = 2 t^1.5
#    (Cha and Mercier, 2019, Example 2): X(t) - X(s) ~ Gamma(A(t) - A(s), b)
x <- sim_gamma(t, b = 3, A = function(t) 2 * t^1.5, n_traj = 10)
x
plot_traj(x, "Non-homogeneous gamma process")

# 6. A(t) = a t gives back the homogeneous process with shape a
set.seed(2)
y1 <- sim_gamma(t, a = 2, b = 3)
set.seed(2)
y2 <- sim_gamma(t, b = 3, A = function(t) 2 * t)
all.equal(unclass(y1)[, 2], unclass(y2)[, 2])

# 7. All the simulated data in a single data frame, with the increments
head(traj_df(x))
head(traj_df(x, increments = TRUE))
```

`sim_ig` *Simulation of the inverse Gaussian process*

Description

Simulates exactly, at given inspection times, one or several trajectories of a homogeneous inverse Gaussian (IG) process (a Lévy process) or of a non-homogeneous IG process, obtained by a time change A.

Usage

```
sim_ig(t, mu, eta, A = NULL, n_traj = 1)
```

Arguments

`t`	Inspection times. The times of each trajectory must start at 0 and be strictly increasing. Three forms are accepted: • a numeric vector, for instance `seq(0, 10, by = 0.5)`: all the trajectories share these times, and their number is `n_traj`; • a list of vectors, one per trajectory, for instance `list(c(0, 1, 3), c(0, 2, 4, 6))`: each trajectory has its own times (`n_traj` is then ignored); • the data frame returned by `inspection_times()`, which gives one plan per trajectory (`n_traj` is then ignored).

`mu`	Mean degradation per unit of time of the homogeneous process, $\mu > 0$. Not given when `A` is given (the mean of $X(t)$ is then $A(t)$).
`eta`	Shape parameter, $\eta > 0$: the larger η, the less variable the increments. The variance of $X(t)$ is $\mu^3 t/\eta$ (homogeneous case) or $A(t)/\eta$ (non-homogeneous case).
`A`	`NULL` (default) for the homogeneous process, or the time change for the non-homogeneous process: a vectorized R function of `t`, non-decreasing with $A(0) = 0$, for instance `function(t) 0.5 * t^1.5`.
`n_traj`	Number of trajectories to simulate when `t` is a vector (default 1), a positive integer.

Value

An object of class `DegradSim`, whose form depends on the inspection times:

- **same times for all the trajectories** (`t` a vector, or a plan of `inspection_times()` with the same times for all the trajectories): a matrix with columns `t` (inspection times), `X1`, ..., `Xn` (simulated degradation levels of the trajectories 1 to n, equal to 0 at `t = 0`). The levels of the k-th trajectory are `x[, k + 1]` or `x[, "Xk"]`;
- **different times** (`t` a list, or a plan of `inspection_times()` with different times): a list with one data frame per trajectory, with columns `t` and `X`. The k-th trajectory is the data frame `x[[k]]`, its times are `x[[k]]$t` (or `x[[k]][, 1]`) and its levels `x[[k]]$X` (or `x[[k]][, 2]`).

The model, the parameters and, for a non-homogeneous process, the time change are stored as attributes. Printing the object shows them with the first simulated values. Use `traj_df()` to gather the data in a single data frame (with the increments if needed) and `plot_traj()` to plot the trajectories.

Model

$IG(m, \lambda)$ denotes the inverse Gaussian distribution with mean $m > 0$ and shape $\lambda > 0$, of density

$$\sqrt{\frac{\lambda}{2\pi x^3}} \exp\Big(-\frac{\lambda(x-m)^2}{2m^2 x}\Big), \qquad x > 0,$$

and variance m^3/λ.

Homogeneous IG process (`A = NULL`, default). Let $\mu > 0$ and $\eta > 0$. The process satisfies $X(0) = 0$, has independent increments, and for $0 \le s < t$,

$$X(t) - X(s) \sim IG\big(\mu(t-s),\ \eta(t-s)^2\big),$$

with the parametrization of Wang and Xu (2010). Then $E[X(t)] = \mu t$ and $Var(X(t)) = \mu^3 t/\eta$. It is a Lévy process, increasing and purely made of jumps.

Non-homogeneous IG process (`A` given). Let A be a time change: a non-decreasing function, right-continuous with left limits, with $A(0) = 0$ and $A(t) \to +\infty$ as $t \to +\infty$. The process is $X(t) = Y(A(t))$, where Y is the homogeneous IG process with $\mu = 1$, so that

$$X(t) - X(s) \sim IG\big(A(t) - A(s),\ \eta\,(A(t) - A(s))^2\big),$$

$E[X(t)] = A(t)$ and $Var(X(t)) = A(t)/\eta$. This is the non-homogeneous IG process with mean function A of Cha and Mercier (2019, Example 3), whose parameter b is η.

Why $\mu = 1$ in the non-homogeneous case. With a mean $\mu\Delta A$, replacing (A, μ, η) by $(cA, \mu/c, \eta/c^2)$, $c > 0$, gives the same distribution $IG(\mu\Delta A, \eta\Delta A^2)$ of the increments: μ, η and A could not be identified separately. The mean is therefore contained in A. With $A(t) = \mu t$, the non-homogeneous process has increments $IG(\mu\Delta t, \eta\mu^2\Delta t^2)$: the homogeneous process with parameters (μ, η) is obtained with $A(t) = \mu t$ and η/μ^2.

Exact simulation at the inspection times (random walk approximation)

The increments over disjoint intervals are independent with known distributions, so the values at the inspection times $0 = t_0 < t_1 < \cdots < t_n$ are simulated exactly by the random walk approximation (RWA) of Kahle et al. (2016, section 2.5.1, written there for the gamma process): independent increments are drawn, then cumulated.

1. $m_i = \mu\Delta t_i$ and $\lambda_i = \eta\Delta t_i^2$ (homogeneous case), or $m_i = \Delta A_i$ and $\lambda_i = \eta\Delta A_i^2$ (non-homogeneous case), with $\Delta t_i = t_i - t_{i-1}$ and $\Delta A_i = A(t_i) - A(t_{i-1})$.
2. Draw independently $\Delta X_i \sim IG(m_i, \lambda_i)$ by the method of Michael, Schucany and Haas (1976), based on $\lambda(X-m)^2/(m^2 X) \sim \chi^2_1$: draw $Z \sim N(0,1)$, set $\nu = Z^2$, $x_1 = m + m^2\nu/(2\lambda) - m\sqrt{4m\lambda\nu + m^2\nu^2}/(2\lambda)$, and return x_1 with probability $m/(m+x_1)$, otherwise m^2/x_1.
3. $X(t_0) = 0$ and $X(t_i) = \sum_{j \le i} \Delta X_j$.

The values at the inspection times have exactly the distribution of the process; only the path between two inspection times is not simulated. If A is constant between two inspection times, the increment is 0.

Conditions on the inspection times and on A

The times of each trajectory must be finite, start at 0 and be strictly increasing, with at least two times (0 and one inspection). The function `A` must be vectorized (`A(t)` returns one value per time), with $A(0) = 0$, finite and non-decreasing at the inspection times: these conditions are checked and an error is raised otherwise. The conditions "right-continuous with left limits" and $A(t) \to +\infty$ cannot be checked on a finite set of times: they are left to the user.

References

Cha, J. H. and Mercier, S. (2019). Transformed Lévy processes as state-dependent wear models. *Advances in Applied Probability*, 51(2), 468-486. doi:10.1017/apr.2019.21

Kahle, W., Mercier, S. and Paroissin, C. (2016). *Degradation Processes in Reliability*. ISTE and Wiley, London. ISBN 978-1-84821-888-8.

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Wang, X. and Xu, D. (2010). An inverse Gaussian process model for degradation data. *Technometrics*, 52(2), 188-197. doi:10.1198/TECH.2009.08197

See Also

`sim_gamma()` and `sim_wiener()` for the other processes with a time change, `plot_traj()`, `traj_df()`, `inspection_times()`, `plot_plan()`

Examples

```
# 1. Homogeneous process: one trajectory inspected every 0.5 time unit
t <- seq(0, 10, by = 0.5)
set.seed(1)
x <- sim_ig(t, mu = 0.5, eta = 2)
x
plot_traj(x)

# 2. Ten trajectories with the same inspection times
x <- sim_ig(t, mu = 0.5, eta = 2, n_traj = 10)
```

```
plot_traj(x)

# Same times for all the trajectories: a matrix with the column t,
# then one column per trajectory (X1, X2, ...)
head(x[, 1:4])
x[, "X2"]

# 3. Irregular inspections, less and less frequent over time
plan <- inspection_times("decreasing", n = 20, n_traj = 5, tmax = 10)
x <- sim_ig(plan, mu = 0.5, eta = 2)
plot_traj(x)

# 4. Different inspection times for each trajectory: a list of data frames
x <- sim_ig(list(c(0, 1, 3, 6), c(0, 2, 4, 6, 8, 10)), mu = 0.5, eta = 2)
x[[2]]
x[[2]]$X

# 5. Non-homogeneous IG process with mean function A(t) = 0.5 t^1.5
#    (Cha and Mercier, 2019, Example 3): X(t) - X(s) ~ IG(dA, eta dA^2)
x <- sim_ig(t, eta = 2, A = function(t) 0.5 * t^1.5, n_traj = 10)
x
plot_traj(x, "Non-homogeneous inverse Gaussian process")

# 6. The homogeneous process (mu, eta) is A(t) = mu t with eta / mu^2
set.seed(2)
y1 <- sim_ig(t, mu = 0.5, eta = 2)
set.seed(2)
y2 <- sim_ig(t, eta = 2 / 0.5^2, A = function(t) 0.5 * t)
all.equal(unclass(y1)[, 2], unclass(y2)[, 2])

# 7. All the simulated data in a single data frame, with the increments
head(traj_df(x))
head(traj_df(x, increments = TRUE))
```

`sim_meixner` *Simulation of the Meixner process*

Description

Simulates exactly, at given inspection times, one or several trajectories of a homogeneous Meixner process, a Lévy process.

Usage

```
sim_meixner(t, a, b, d, m, n_traj = 1)
```

Arguments

`t`	Inspection times. The times of each trajectory must start at 0 and be strictly increasing. Three forms are accepted: • a numeric vector, for instance `seq(0, 10, by = 0.5)`: all the trajectories share these times, and their number is `n_traj`; • a list of vectors, one per trajectory, for instance `list(c(0, 1, 3), c(0, 2, 4, 6))`: each trajectory has its own times (`n_traj` is then ignored);

- the data frame returned by `inspection_times()`, which gives one plan per trajectory (`n_traj` is then ignored).

`a` Scale parameter, $a > 0$.

`b` Asymmetry parameter, $-\pi < b < \pi$: $b = 0$ gives symmetric increments, $b > 0$ increments skewed to the right.

`d` Shape parameter per unit of time, $d > 0$.

`m` Location parameter per unit of time, real.

`n_traj` Number of trajectories to simulate when `t` is a vector (default 1), a positive integer.

Value

An object of class `DegradSim`, whose form depends on the inspection times:

- **same times for all the trajectories** (`t` a vector, or a plan of `inspection_times()` with the same times for all the trajectories): a matrix with columns `t` (inspection times), `X1`, ..., `Xn` (simulated degradation levels of the trajectories 1 to n, equal to 0 at `t = 0`). The levels of the k-th trajectory are `x[, k + 1]` or `x[, "Xk"]`;
- **different times** (`t` a list, or a plan of `inspection_times()` with different times): a list with one data frame per trajectory, with columns `t` and `X`. The k-th trajectory is the data frame `x[[k]]`, its times are `x[[k]]$t` (or `x[[k]][, 1]`) and its levels `x[[k]]$X` (or `x[[k]][, 2]`).

The model, the parameters and, for a non-homogeneous process, the time change are stored as attributes. Printing the object shows them with the first simulated values. Use `traj_df()` to gather the data in a single data frame (with the increments if needed) and `plot_traj()` to plot the trajectories.

Model

The $Meixner(a, b, d, m)$ distribution, with $a > 0$, $b \in]-\pi, \pi[$, $d > 0$ and m real, has density

$$f(x) = \frac{(2\cos(b/2))^{2d}}{2a\pi\,\Gamma(2d)} \exp\Big(\frac{b(x-m)}{a}\Big)\Big|\Gamma\Big(d + i\frac{x-m}{a}\Big)\Big|^2, \qquad x \in \mathbb{R}.$$

It is infinitely divisible: the sum of independent $Meixner(a, b, d_1, m_1)$ and $Meixner(a, b, d_2, m_2)$ variables has the $Meixner(a, b, d_1 + d_2, m_1 + m_2)$ distribution.

The Meixner process satisfies $X(0) = 0$, has independent increments, and for $0 \le s < t$,

$$X(t) - X(s) \sim Meixner\big(a, b, d(t-s), m(t-s)\big).$$

Hence $E[X(t)] = (m + ad\tan(b/2))\,t$ and $Var(X(t)) = a^2 d\,t/(2\cos^2(b/2))$. It is a Lévy process whose trajectories are not monotone, purely made of jumps, of infinite variation: a is a scale parameter, b sets the asymmetry, d the shape and m the location.

Exact simulation at the inspection times (random walk approximation)

The increments over disjoint intervals are independent with known distributions, so the values at the inspection times $0 = t_0 < t_1 < \cdots < t_n$ are simulated by the random walk approximation (RWA) of Kahle et al. (2016, section 2.5.1, written there for the gamma process):

1. draw independently $\Delta X_i \sim Meixner(a, b, d\Delta t_i, m\Delta t_i)$, with $\Delta t_i = t_i - t_{i-1}$;
2. $X(t_0) = 0$ and $X(t_i) = \sum_{j \le i} \Delta X_j$.

There is no Meixner generator in `stats`: each increment is drawn by numerical inversion of its distribution function (method PINV of the package `Runuran`, built from the exact density, with an error in u of order 10^{-10}). One generator is built for each distinct time step Δt_i and reused for all the trajectories. The simulation is therefore exact up to this numerical error.

Conditions on the inspection times

The times of each trajectory must be finite, start at 0 and be strictly increasing, with at least two times (0 and one inspection); an error is raised otherwise.

References

Kahle, W., Mercier, S. and Paroissin, C. (2016). *Degradation Processes in Reliability*. ISTE and Wiley, London. ISBN 978-1-84821-888-8.

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Hörmann, W. and Leydold, J. (2003). Continuous random variate generation by fast numerical inversion. *ACM Transactions on Modeling and Computer Simulation*, 13(4), 347-362. doi:10.1145/945511.945517

See Also

`plot_traj()`, `traj_df()`, `inspection_times()`, `plot_plan()`

Examples

```
# 1. One trajectory inspected every 0.5 time unit up to 10
t <- seq(0, 10, by = 0.5)
set.seed(1)
x <- sim_meixner(t, a = 1, b = 0.3, d = 1.5, m = 0.2)
x
plot_traj(x)

# 2. Ten trajectories with the same inspection times
x <- sim_meixner(t, a = 1, b = 0.3, d = 1.5, m = 0.2, n_traj = 10)
plot_traj(x)

# Same times for all the trajectories: a matrix with the column t,
# then one column per trajectory (X1, X2, ...)
head(x[, 1:4])
x[, "X2"]

# 3. Irregular inspections, frequent in the middle of the period
plan <- inspection_times("inc-dec", n = 20, n_traj = 5, tmax = 10)
x <- sim_meixner(plan, a = 1, b = 0.3, d = 1.5, m = 0.2)
plot_traj(x)

# 4. Different inspection times for each trajectory: a list of data frames
x <- sim_meixner(list(c(0, 1, 3, 6), c(0, 2, 4, 6, 8, 10)), a = 1, b = 0.3, d = 1.5, m = 0.2)
x[[2]]
x[[2]]$X

# 5. Symmetric increments (b = 0) and no drift (m = 0)
x <- sim_meixner(t, a = 1, b = 0, d = 1.5, m = 0, n_traj = 10)
plot_traj(x, "Symmetric Meixner process")

# 6. All the simulated data in a single data frame, with the increments
head(traj_df(x))
head(traj_df(x, increments = TRUE))
```

`sim_vg` *Simulation of the variance-gamma process*

Description

Simulates exactly, at given inspection times, one or several trajectories of a homogeneous variance-gamma (VG) process, a Lévy process.

Usage

```
sim_vg(t, theta, sigma, nu, method = "subordination", n_traj = 1)
```

Arguments

`t`	Inspection times. The times of each trajectory must start at 0 and be strictly increasing. Three forms are accepted: • a numeric vector, for instance `seq(0, 10, by = 0.5)`: all the trajectories share these times, and their number is `n_traj`; • a list of vectors, one per trajectory, for instance `list(c(0, 1, 3), c(0, 2, 4, 6))`: each trajectory has its own times (`n_traj` is then ignored); • the data frame returned by `inspection_times()`, which gives one plan per trajectory (`n_traj` is then ignored).
`theta`	Drift of the subordinated Brownian motion, real: mean degradation per unit of time.
`sigma`	Diffusion coefficient of the subordinated Brownian motion, $\sigma > 0$ (standard deviation, not variance).
`nu`	Variance rate of the gamma clock, $\nu > 0$: the larger ν, the larger the jumps; as $\nu \to 0$ the process becomes a Wiener process.
`method`	Simulation algorithm: `"subordination"` (default) or `"difference"` (difference of two gamma processes). Both are exact and give the same distribution; see the section on the exact simulation.
`n_traj`	Number of trajectories to simulate when `t` is a vector (default 1), a positive integer.

Value

An object of class `DegradSim`, whose form depends on the inspection times:

- **same times for all the trajectories** (`t` a vector, or a plan of `inspection_times()` with the same times for all the trajectories): a matrix with columns `t` (inspection times), `X1`, ..., `Xn` (simulated degradation levels of the trajectories 1 to n, equal to 0 at `t = 0`). The levels of the k-th trajectory are `x[, k + 1]` or `x[, "Xk"]`;
- **different times** (`t` a list, or a plan of `inspection_times()` with different times): a list with one data frame per trajectory, with columns `t` and `X`. The k-th trajectory is the data frame `x[[k]]`, its times are `x[[k]]$t` (or `x[[k]][, 1]`) and its levels `x[[k]]$X` (or `x[[k]][, 2]`).

The model, the parameters and, for a non-homogeneous process, the time change are stored as attributes. Printing the object shows them with the first simulated values. Use `traj_df()` to gather the data in a single data frame (with the increments if needed) and `plot_traj()` to plot the trajectories.

Model

Let θ be real, $\sigma > 0$ and $\nu > 0$. Let G be a gamma process with $G(t) - G(s) \sim \Gamma((t-s)/\nu, \text{rate } 1/\nu)$, so that $E[G(t)] = t$ and $Var(G(t)) = \nu t$, and B an independent standard Brownian motion. The VG process is the Brownian motion with drift evaluated at the random clock G:

$$X(t) = \theta G(t) + \sigma B(G(t)), \qquad t \geq 0.$$

It is a Lévy process: $X(0) = 0$, its increments are independent and stationary, $E[X(t)] = \theta t$ and $Var(X(t)) = (\sigma^2 + \theta^2\nu)\, t$. The trajectories are not monotone, purely made of jumps, of finite variation. As $\nu \to 0$, the VG process tends to the Wiener process $\theta t + \sigma B(t)$.

Exact simulation at the inspection times (random walk approximation)

The increments over disjoint intervals are independent with known distributions, so the values at the inspection times $0 = t_0 < t_1 < \cdots < t_n$ are simulated exactly by the random walk approximation (RWA) of Kahle et al. (2016, section 2.5.1, written there for the gamma process): independent increments are drawn, then cumulated. Each increment, over a time $\Delta t_i = t_i - t_{i-1}$, is drawn by one of two exact methods (argument `method`), which give the same distribution.

Method 1, subordination (`method = "subordination"`, default). Conditionally on the clock increment ΔG, $X(t) - X(s) \sim N(\theta\Delta G, \sigma^2\Delta G)$:

1. draw independently $\Delta G_i \sim \Gamma(\Delta t_i/\nu)$ with scale ν;
2. $\Delta X_i = \theta\Delta G_i + \sigma\sqrt{\Delta G_i}\, Z_i$, with Z_i independent $N(0,1)$;
3. $X(t_0) = 0$ and $X(t_i) = \sum_{j \leq i} \Delta X_j$.

This method follows the definition of the process and also gives the random clock G.

Method 2, difference of two gamma processes (`method = "difference"`). Let

$$\mu_\pm = \frac{1}{2}\sqrt{\theta^2 + \frac{2\sigma^2}{\nu}} \pm \frac{\theta}{2} > 0,$$

and G_+, G_- two independent gamma processes with shapes t/ν and scales $\nu\mu_\pm$. Then X has the same distribution as $G_+ - G_-$: since $\mu_+ - \mu_- = \theta$ and $\nu^2\mu_+\mu_- = \sigma^2\nu/2$, the characteristic function of $\Delta G_+ - \Delta G_-$ over a time Δt is

$$(1 - iu\nu\mu_+)^{-\Delta t/\nu}(1 + iu\nu\mu_-)^{-\Delta t/\nu} = \left(1 - iu\theta\nu + \sigma^2\nu u^2/2\right)^{-\Delta t/\nu},$$

which is the characteristic function of the VG increment (Madan et al., 1998).

1. Draw independently $\Delta G_i^+ \sim \Gamma(\Delta t_i/\nu)$ with scale $\nu\mu_+$ and $\Delta G_i^- \sim \Gamma(\Delta t_i/\nu)$ with scale $\nu\mu_-$;
2. $\Delta X_i = \Delta G_i^+ - \Delta G_i^-$;
3. $X(t_0) = 0$ and $X(t_i) = \sum_{j \leq i} \Delta X_j$.

This method only uses gamma draws.

The values at the inspection times have exactly the distribution of the process; only the path between two inspection times is not simulated.

Conditions on the inspection times

The times of each trajectory must be finite, start at 0 and be strictly increasing, with at least two times (0 and one inspection); an error is raised otherwise.

References

Kahle, W., Mercier, S. and Paroissin, C. (2016). *Degradation Processes in Reliability*. ISTE and Wiley, London. ISBN 978-1-84821-888-8.

Madan, D. B., Carr, P. P. and Chang, E. C. (1998). The variance gamma process and option pricing. *European Finance Review*, 2(1), 79-105. doi:10.1023/A:1009703431535

See Also

`plot_traj()`, `traj_df()`, `inspection_times()`, `plot_plan()`

Examples

```
# 1. One trajectory inspected every 0.5 time unit up to 10
t <- seq(0, 10, by = 0.5)
set.seed(1)
x <- sim_vg(t, theta = 0.5, sigma = 1.2, nu = 0.8)
x
plot_traj(x)

# 2. Ten trajectories, simulated by subordination (default)
x <- sim_vg(t, theta = 0.5, sigma = 1.2, nu = 0.8, n_traj = 10)
plot_traj(x)

# Same times for all the trajectories: a matrix with the column t,
# then one column per trajectory (X1, X2, ...)
head(x[, 1:4])
x[, "X2"]

# 3. The same model simulated as a difference of two gamma processes
y <- sim_vg(t, theta = 0.5, sigma = 1.2, nu = 0.8, method = "difference", n_traj = 10)
plot_traj(y, "Variance-gamma process, difference of two gamma processes")

# 4. Irregular inspections, more and more frequent over time
plan <- inspection_times("increasing", n = 20, n_traj = 5, tmax = 10)
x <- sim_vg(plan, theta = 0.5, sigma = 1.2, nu = 0.8)
plot_traj(x)

# 5. Different inspection times for each trajectory: a list of data frames
x <- sim_vg(list(c(0, 1, 3, 6), c(0, 2, 4, 6, 8, 10)), theta = 0.5, sigma = 1.2, nu = 0.8)
x[[2]]
x[[2]]$X

# 6. Small nu: the trajectories look like those of a Wiener process
x <- sim_vg(t, theta = 0.5, sigma = 1.2, nu = 0.01, n_traj = 10)
plot_traj(x, "Variance-gamma process with nu = 0.01")

# 7. All the simulated data in a single data frame, with the increments
head(traj_df(x))
head(traj_df(x, increments = TRUE))
```

`sim_wiener` *Simulation of the Wiener process*

Description

Simulates exactly, at given inspection times, one or several trajectories of a homogeneous Wiener process with drift (a Lévy process) or of a time-scaled Wiener process, obtained by a time change A.

Usage

```
sim_wiener(t, mu, sigma, A = NULL, n_traj = 1)
```

Arguments

`t`	Inspection times. The times of each trajectory must start at 0 and be strictly increasing. Three forms are accepted: • a numeric vector, for instance `seq(0, 10, by = 0.5)`: all the trajectories share these times, and their number is `n_traj`; • a list of vectors, one per trajectory, for instance `list(c(0, 1, 3), c(0, 2, 4, 6))`: each trajectory has its own times (`n_traj` is then ignored); • the data frame returned by `inspection_times()`, which gives one plan per trajectory (`n_traj` is then ignored).
`mu`	Drift of the homogeneous process, real: mean degradation per unit of time. Not given when `A` is given (the mean of $X(t)$ is then $A(t)$).
`sigma`	Diffusion coefficient, $\sigma > 0$: standard deviation (not variance) of $X(t) - X(s)$ for $t - s = 1$ (homogeneous case) or $A(t) - A(s) = 1$ (time-scaled case).
`A`	`NULL` (default) for the homogeneous process, or the time change for the time-scaled process: a vectorized R function of `t`, non-decreasing with $A(0) = 0$, for instance `function(t) 0.5 * t^1.5`.
`n_traj`	Number of trajectories to simulate when `t` is a vector (default 1), a positive integer.

Value

An object of class `DegradSim`, whose form depends on the inspection times:

- **same times for all the trajectories** (`t` a vector, or a plan of `inspection_times()` with the same times for all the trajectories): a matrix with columns `t` (inspection times), `X1`, ..., `Xn` (simulated degradation levels of the trajectories 1 to n, equal to 0 at `t = 0`). The levels of the k-th trajectory are `x[, k + 1]` or `x[, "Xk"]`;
- **different times** (`t` a list, or a plan of `inspection_times()` with different times): a list with one data frame per trajectory, with columns `t` and `X`. The k-th trajectory is the data frame `x[[k]]`, its times are `x[[k]]$t` (or `x[[k]][, 1]`) and its levels `x[[k]]$X` (or `x[[k]][, 2]`).

The model, the parameters and, for a non-homogeneous process, the time change are stored as attributes. Printing the object shows them with the first simulated values. Use `traj_df()` to gather the data in a single data frame (with the increments if needed) and `plot_traj()` to plot the trajectories.

Model

B denotes a standard Brownian motion and $N(m, v)$ the normal distribution with mean m and variance v.

Homogeneous Wiener process (`A = NULL`, default). Let μ be real and $\sigma > 0$. The process is

$$X(t) = \mu t + \sigma B(t), \qquad t \geq 0,$$

where μ is the drift parameter (expected to be positive for a degradation) and σ^2 the variance (or volatility) parameter. It has independent increments $X(t) - X(s) \sim N(\mu(t-s), \sigma^2(t-s))$, $E[X(t)] = \mu t$ and $Var(X(t)) = \sigma^2 t$. It is a Lévy process with continuous, non-monotone trajectories.

Time-scaled Wiener process (`A` given). Let A be a time change: a non-decreasing function, right-continuous with left limits, with $A(0) = 0$ and $A(t) \to +\infty$ as $t \to +\infty$. The process is

$$X(t) = A(t) + \sigma B(A(t)), \qquad t \geq 0,$$

that is $X(t) = Y(A(t))$ with $Y(u) = u + \sigma B(u)$ (Cha and Mercier, 2019, Example 1, with $\mu = 1$). Its increments are independent, $X(t) - X(s) \sim N(A(t) - A(s), \sigma^2(A(t) - A(s)))$, $E[X(t)] = A(t)$ and $Var(X(t)) = \sigma^2 A(t)$.

Why $\mu = 1$ in the time-scaled case. With a drift μ, the processes $\mu A(t) + \sigma B(A(t))$ and $(\mu/c)(cA(t)) + (\sigma/\sqrt{c})B(cA(t))$ have the same distribution for every $c > 0$: μ, σ and A could not be identified separately. The drift is therefore contained in A. With $A(t) = \mu t$ ($\mu > 0$), the time-scaled process is the homogeneous process with drift μ and variance parameter $\sigma^2\mu$: the homogeneous process with parameters (μ, σ) is obtained with $A(t) = \mu t$ and $\sigma/\sqrt{\mu}$.

Exact simulation at the inspection times (random walk approximation)

The increments over disjoint intervals are independent normal variables, so the values at the inspection times $0 = t_0 < t_1 < \cdots < t_n$ are simulated exactly by the random walk approximation (RWA) of Kahle et al. (2016, section 1.3.1, algorithm 1.7):

1. $\Delta X_i = \mu \Delta t_i + \sigma\sqrt{\Delta t_i}\, Z_i$ (homogeneous case) or $\Delta X_i = \Delta A_i + \sigma\sqrt{\Delta A_i}\, Z_i$ (time-scaled case), with $\Delta t_i = t_i - t_{i-1}$, $\Delta A_i = A(t_i) - A(t_{i-1})$ and Z_i independent $N(0,1)$ drawn with `stats::rnorm()`;
2. $X(t_0) = 0$ and $X(t_i) = \sum_{j \leq i} \Delta X_j$.

The values at the inspection times have exactly the distribution of the process; only the path between two inspection times is not simulated.

Conditions on the inspection times and on A

The times of each trajectory must be finite, start at 0 and be strictly increasing, with at least two times (0 and one inspection). The function `A` must be vectorized (`A(t)` returns one value per time), with $A(0) = 0$, finite and non-decreasing at the inspection times: these conditions are checked and an error is raised otherwise. The conditions "right-continuous with left limits" and $A(t) \to +\infty$ cannot be checked on a finite set of times: they are left to the user.

References

Kahle, W., Mercier, S. and Paroissin, C. (2016). *Degradation Processes in Reliability*. ISTE and Wiley, London. ISBN 978-1-84821-888-8.

Cha, J. H. and Mercier, S. (2019). Transformed Lévy processes as state-dependent wear models. *Advances in Applied Probability*, 51(2), 468-486. doi:10.1017/apr.2019.21

Whitmore, G. A. (1995). Estimating degradation by a Wiener diffusion process subject to measurement error. *Lifetime Data Analysis*, 1, 307-319.

See Also

`sim_gamma()` and `sim_ig()` for the other processes with a time change, `plot_traj()`, `traj_df()`, `inspection_times()`, `plot_plan()`

Examples

```
# 1. Homogeneous process: one trajectory inspected every 0.5 time unit
t <- seq(0, 10, by = 0.5)
set.seed(1)
x <- sim_wiener(t, mu = 0.5, sigma = 1.5)
x
plot_traj(x)

# 2. Ten trajectories with the same inspection times
x <- sim_wiener(t, mu = 0.5, sigma = 1.5, n_traj = 10)
plot_traj(x)

# Same times for all the trajectories: a matrix with the column t,
# then one column per trajectory (X1, X2, ...)
head(x[, 1:4])
x[, "X2"]

# 3. Irregular inspections, frequent at the beginning and at the end
plan <- inspection_times("dec-inc", n = 20, n_traj = 5, tmax = 10)
x <- sim_wiener(plan, mu = 0.5, sigma = 1.5)
plot_traj(x)

# 4. Different inspection times for each trajectory: a list of data frames
x <- sim_wiener(list(c(0, 1, 3, 6), c(0, 2, 4, 6, 8, 10)), mu = 0.5, sigma = 1.5)
x[[2]]
x[[2]]$X

# 5. Time-scaled Wiener process X(t) = A(t) + sigma B(A(t)) with
#    A(t) = 0.5 t^1.5 (Cha and Mercier, 2019, Example 1)
x <- sim_wiener(t, sigma = 1.5, A = function(t) 0.5 * t^1.5, n_traj = 10)
x
plot_traj(x, "Time-scaled Wiener process")

# 6. The homogeneous process (mu, sigma) is A(t) = mu t with sigma / sqrt(mu)
set.seed(2)
y1 <- sim_wiener(t, mu = 0.5, sigma = 1.5)
set.seed(2)
y2 <- sim_wiener(t, sigma = 1.5 / sqrt(0.5), A = function(t) 0.5 * t)
all.equal(unclass(y1)[, 2], unclass(y2)[, 2])

# 7. All the simulated data in a single data frame, with the increments
head(traj_df(x))
head(traj_df(x, increments = TRUE))
```

`traj_df` *Simulated data as a data frame*

Description

Gathers the trajectories simulated by a `sim_` function into a single data frame, one row per inspection, ready to be saved, summarized or used by other functions. Optionally adds the time steps and the increments between two consecutive inspections.

Usage

```
traj_df(x, increments = FALSE)
```

Arguments

x	Result of a simulation function of the package, for instance sim_gamma().
increments	FALSE (default) for the levels only; TRUE to add the columns dt and dX.

Value

A data frame with one row per inspection and the columns:

- id: number of the trajectory (1, 2, ...);
- t: inspection time;
- X: simulated degradation level at time t;
- with increments = TRUE, also dt, the time elapsed since the previous inspection, and dX, the increment of the level since the previous inspection (NA at the first inspection, t = 0, of each trajectory).

To work on a single trajectory, use subset(traj_df(x), id == k).

See Also

plot_traj()

Examples

```
set.seed(1)
x <- sim_gamma(seq(0, 10, by = 1), a = 2, b = 3, n_traj = 3)
d <- traj_df(x)
head(d)

# Degradation level of each trajectory at the last inspection
d[d$t == 10, ]

# Time steps and increments between two inspections
head(traj_df(x, increments = TRUE))

# Trajectories with different inspection times
y <- sim_gamma(list(c(0, 1, 3, 6), c(0, 2, 4, 6, 8)), a = 2, b = 3)
traj_df(y, increments = TRUE)

# Save the simulated data in a csv file
f <- tempfile(fileext = ".csv")
write.csv(d, f, row.names = FALSE)
```

Index

∗ **package**
DegradSim-package, 2

DegradSim *(*DegradSim-package*)*, 2
DegradSim-package, 2

inspection_times, 6
inspection_times(), *2–4*, *9–15*, *17–19*, *21–23*

plot_plan, 8
plot_plan(), *4*, *6, 7*, *12*, *15*, *18*, *21*, *23*
plot_traj, 9
plot_traj(), *4*, *9*, *11, 12*, *14, 15*, *17–19*, *21–23*, *25*

sim_gamma, 10
sim_gamma(), *2, 3*, *6, 7*, *10*, *15*, *23*, *25*
sim_ig, 13
sim_ig(), *2, 3*, *12*, *23*
sim_meixner, 16
sim_meixner(), *2*
sim_vg, 19
sim_vg(), *2*
sim_wiener, 21
sim_wiener(), *2, 3*, *12*, *15*
stats::rgamma(), *4*, *12*
stats::rnorm(), *4*, *23*

traj_df, 24
traj_df(), *4*, *10–12*, *14, 15*, *17–19*, *21–23*